

MID SEMESTER EXAMINATION 2025

Name of the Program: B.Tech

Name of the Course: Programming for Problem solving

Course Code: TCS201

Time: 90 Minutes

Maximum Marks: 50

Semester: II

Note:

- (i) Answer all the questions except Que(2) by choosing any one of the sub-questions.
 (ii) All 5 sub-questions in Que(2) are compulsory.
 (iii) Each question carries 10 marks.

Q1	(10 marks)	Draw a flow-chart to input elements into a matrix of size $m \times n$. Find sum and average of the matrix elements. Also count the number of elements greater than the average.	
OR			
(b)		Draw a flowchart to read a multiword string and print initial letter of each word in upper case.	CO2
Sample Input:		Sample Output:	
Inputted String: Aditya birla group		A B G	
Inputted String: State bank Of India		S B O I	
Q2	(10 marks)	Find the Output with an explanation. Considering <code>int</code> is getting 2 Bytes of space in memory.	
1.		<pre>#include<stdio.h> int main() { int a[20] = {20, 60, 100, 140}; int *p = a; for (i=0; i<4; i++) { *p = *p + 5; p = p + 1; *p = 100; printf("%d", a[i]); } return 0; }</pre>	CO1 CO3
2.		<pre>#include<stdio.h> void changeValue(int *p) { *p = 20; } int main() { int a = 10; changeValue(&a); printf("%d", a); return 0; }</pre>	
3.		<pre>#include<stdio.h> int main() { if (printf("Hello") == strlen("Hello")) printf("...Friends"); else printf("...Enemies"); }</pre>	
4.		<pre>#include<stdio.h> int main() { char ch[] = "GOD"; int x = strlen(ch); int y = sizeof(ch); }</pre>	

		<pre>return 0; } printf("%d %d", x, y); }</pre>	
Q3	(10 marks)	5. <pre>#include<stdio.h> int main() { int arr[] = {10, 20, 30, 40}; int *ptr = arr; printf("%d", *(ptr + 2)); return 0; }</pre>	
(a)		Read elements for a 2D array of size $m \times n$. Write a 'C' code to sort each row of matrix and print the final matrix.	
Sample Input:		Sample Output:	
Number of rows: 3		Final matrix:	
Number of columns: 5		16 23 34 45 67	
Elements of matrix:		12 27 36 76 81	
23 16 34 67 45		11 21 72 89 90	
81 36 12 27 76			
72 89 90 11 21			
OR			
(b)		Mr. X a mathematician deals with the problems of matrices. To solve a problem, he needs identity matrix. So, he inputted elements for a square matrix of size $m \times n$. Write a code in 'C' to help him to identify whether the matrix is identity matrix or not. Note: Identity matrix is matrix whose principal diagonal elements are 1 and other elements are zero.	CO1
Sample Input:		Sample Output:	
Size of matrix: 3		It is an Identity Matrix	
Elements of matrix:			
1 0 0			
0 1 0			
0 0 1			
Size of matrix: 3		It is not an Identity Matrix	
Elements of matrix:			
1 0 1			
0 1 0			
2 0 1			
Q4	(10 marks)	Differentiate string with character array. Describe any two strings handling library function. Write a 'C' program to read a multiword string. Input a character and find first occurrence of that character in the string.	
Sample Input:		Sample Output:	
Inputted String: I love my country		First occurrence of character is at	
Inputted Character: o		Index: 3	CO1
OR			

(b)	Differentiate the method of reading string via <code>scanf()</code> and via <code>gets()</code>. Write a 'C' program to read a string. Replace all the vowels according to the criteria as replace <i>a</i> with <i>e</i>, <i>e</i> with <i>i</i>, <i>i</i> with <i>o</i>, <i>o</i> with <i>u</i> and <i>u</i> with <i>a</i>. Print the final string.						
Q5	<table border="1"> <tr> <td data-bbox="173 1655 221 2036">Sample Input:</td> <td data-bbox="173 1285 221 1655">Sample Output:</td> </tr> <tr> <td data-bbox="221 1655 325 2036">Inputted String: Dehradun is a beautiful valley in uttarakhand</td> <td data-bbox="221 1285 325 1655">Final Updated String: Dihredan os e bieatofal vellyi on atterekhend</td> </tr> </table>			Sample Input:	Sample Output:	Inputted String: Dehradun is a beautiful valley in uttarakhand	Final Updated String: Dihredan os e bieatofal vellyi on atterekhend
Sample Input:	Sample Output:						
Inputted String: Dehradun is a beautiful valley in uttarakhand	Final Updated String: Dihredan os e bieatofal vellyi on atterekhend						
(a)	What is a pointer? Explain its advantages. What will be the size of an integer pointer, a character pointer and a float pointer? Write a 'C' program to pass a 1D array of <i>n</i> integer elements (where <i>n</i> is an even number) in a function using pointer as an argument. Swap the adjacent elements and print the final array. <table border="1"> <tr> <td data-bbox="453 1655 499 2036">Sample Input:</td> <td data-bbox="453 1285 499 1655">Sample Output:</td> </tr> <tr> <td data-bbox="499 1655 572 2036">Inputted Array: 10 20 30 40 50 60 70 80</td> <td data-bbox="499 1285 572 1655">Output Array: 20 10 40 30 60 50 80 70</td> </tr> </table>			Sample Input:	Sample Output:	Inputted Array: 10 20 30 40 50 60 70 80	Output Array: 20 10 40 30 60 50 80 70
Sample Input:	Sample Output:						
Inputted Array: 10 20 30 40 50 60 70 80	Output Array: 20 10 40 30 60 50 80 70						
(b)	What do you mean by function returning multiple values? How we can achieve this in 'C' language. Write a 'C' program to pass marks in five subjects of a student into a function (consider maximum marks per subject is 100) and function will return percentage marks and grade of student. Grade Criteria: Percentage >75: Grade 'A' Percentage >60: Grade 'B' Percentage >45: Grade 'C' Otherwise: Grade 'F'						